

Lecture 4 — Networking Devices, Topologies and the Cloud Network

0. Where we left off

Lecture 3 established the OSI and TCP/IP models: layering, encapsulation, and the seven layers from Physical up to Application. We noted then that people say "Layer 3" far more often than they say "Network layer".

This lecture is where that pays off. Almost every device below is introduced by naming the layer it operates at, and the differences between them are almost entirely differences of how deep into the packet they are willing to look.

1. The why of network devices

1.1 Connecting computers directly does not scale

Suppose you want every computer to be able to reach every other computer, and you connect each pair with its own cable. The number of cables you need for n devices is:

$$n(n-1)/2$$

Devices	Calculation	Cables needed
3	$3 \times 2 / 2$	3
10	$10 \times 9 / 2$	45
100	$100 \times 99 / 2$	4,950
1,000	$1,000 \times 999 / 2$	499,500

The last two rows were left for you to work out in class. Notice the shape of the growth: the number of devices grows linearly while the number of cables grows with the square. Ten times the devices costs roughly a hundred times the cabling. There is no version of this that survives a real office, let alone a data centre.

1.2 The idea that fixes it

THE PIVOT

Instead of connecting everyone to everyone — what if everyone connects to ONE device?

Now n devices need n cables, not $n(n-1)/2$. This single move is the reason networking hardware exists at all. Everything that follows in this lecture is a refinement of the question: what should that one device in the middle actually be?

2. Networking devices

These are the core components of a computer network: the things that connect computers and other devices together. The ones named in class were the hub, the switch, the wireless router and the access point, and we then added the firewall and the load balancer.

2.1 The hub — the 1980s answer

- Function: broadcasts incoming data to all connected devices.
- It does not recognise hardware addresses (MAC addresses) at all.
- It operates at the Physical layer — Layer 1.

A hub is the most literal possible interpretation of "connect everyone to one device". It takes the electrical signal arriving on one port and repeats it out of every other port. It has no idea what a device is, let alone which one you meant.

THE QUESTION ASKED IN CLASS

What is the main real-world problem with using a hub in a network with many devices?

Answer: unnecessary traffic. Data is broadcast to every device even when only one device needs it. Every machine on the network has to receive, inspect and discard traffic that was never meant for it — and while that traffic occupies the shared medium, nobody else can transmit. The more devices you add, the worse it gets.

2.2 The switch — the 1990s answer

- Operates at the Data Link layer — Layer 2.
- Connects devices within a LAN and forwards data to the correct device using MAC addresses.
- It has no understanding of Layer 3 — IP addresses mean nothing to it.
- Ports can be configured to control how connections are routed.
- Broadcasting can be controlled, because the switch has MAC-level intelligence about which device sits on which port.

The switch is the hub with a memory. Because it learns which MAC address lives behind which port, it can send a frame only where it needs to go. The unnecessary traffic problem largely disappears.

2.3 Hub versus switch

	Hub	Switch
Era	1980s	1990s onward
Layer	Layer 1 — Physical	Layer 2 — Data Link
Addresses understood	None	MAC addresses
Forwarding behaviour	Broadcasts to every port	Sends to the correct port only
Main weakness	Unnecessary traffic; congestion grows with device count	Cannot see IP, so cannot reach another network

2.4 The router — reaching another network

The switch solves communication inside one network. That immediately raises the next question, which is the one posed in class:

What if we want to communicate with a computer in **another network**?

The switch cannot help, because it has no understanding of Layer 3. The device that can is the router.

- It routes data between different networks.
- It operates at the Network layer — Layer 3.

This is exactly the GPS analogy from Lecture 3: the router is the device that knows about logical addressing and about routes to destinations beyond the local link.

2.5 The firewall — should every packet be allowed in?

A router knows where to send packets. But that raises a different kind of question, and it is a security question rather than a delivery one: should every packet be allowed to enter?

- **A firewall is a network security device.**
- It monitors data coming into and going out of the network.
- Its main job is to protect the network from unauthorised access or threats — only authorised data is allowed through.

Packet filtering

Consider a packet arriving from an external network, bound for the internal network. It passes the router, and then the firewall inspects it. A packet has structure:

Part of the packet	What the firewall checks
IP header	Source and destination IP address
TCP / UDP header	Source and destination port number
Data	Not examined by simple packet filtering

So basic packet filtering is a decision made on four values — where it came from, where it is going, and which ports are involved at each end. Note the layers: the IP address is Layer 3 and the port number is Layer 4. A packet-filtering firewall is a device that reads two layers deep and stops there.

2.6 Load balancers

A load balancer distributes incoming network traffic across multiple servers, to prevent any single server from being overwhelmed. In the diagram from class, six requests arriving from clients were spread across three servers rather than piling onto one.

Major companies — Netflix, Google, Amazon, Facebook — all rely on load balancers to stay fast and smooth even at very high user volumes.

THE LINK BACK TO LECTURE 2

This is the bottleneck idea again, seen from the other side. In Lecture 2 we said end-to-end throughput is capped by the slowest hop. A load balancer is what you build when the slowest hop is a server rather than a link: instead of making one server faster, you stop sending it everything.

2.7 The devices, by layer

This table is the single most useful thing to memorise from this lecture.

Device	Layer	What it makes decisions on
Hub	L1 — Physical	Nothing. It repeats signals to every port.
Switch	L2 — Data Link	MAC addresses, within one LAN.
Router	L3 — Network	IP addresses, between different networks.
Firewall	L3 / L4 (and above)	IP addresses and port numbers; higher types inspect more.
Load balancer	L4 / L7	Which of several servers should receive this request.
Wireless router / access point	L1–L3 depending on role	Named in class as core components; provide wireless access to the network.

3. Network topologies

The devices are ready. That leaves the question of how to arrange them.

Network topology describes the arrangement of devices and their connections within a network, including how data flows through it.

The types named in class were star, mesh, hybrid, ring and bus.

3.1 Star topology

- All devices are connected to a central hub or switch.
- Data flows from the sending device to the hub or switch, which then forwards it to the intended receiver.
- The centralised structure simplifies network diagnostics and maintenance — there is one obvious place to look.
- It is a popular choice for LANs, because the design stays manageable as it grows.

This is the direct descendant of Section 1.2: everyone connects to one device. Its weakness is the mirror image of its strength — that central device is a single point of failure.

3.2 Mesh topology

- Every network device is directly connected to all other devices.
- This creates many possible paths for sending data between any two devices.
- Reliability increases: if one path fails, data can take another route.

You have already met the cost of this arrangement — it is the $n(n-1)/2$ problem from Section 1.1. Mesh topologies are jokingly called spaghetti networks for exactly that reason. The idea remains valuable where the links are not physical cables: wireless mesh networks are widely used to extend WiFi coverage, because adding a node adds paths rather than cabling.

3.3 Hybrid topology

A hybrid topology combines two or more different topologies to form a more complex and more flexible network — for example star, ring and bus segments joined together. Real networks of any size are almost always hybrids, because different parts of an organisation have different needs.

3.5 Choosing a topology by environment

Environment	Suitable topology	Reason	Example devices
Small office	Star	Easy setup, centralised control	Laptops, printers, routers
Smart home	Mesh	Reliable, direct device-to-device communication	Smart bulbs, cameras
Data centre	Hybrid	Combines the benefits of multiple topologies	Servers, storage units

3.6 Fault tolerance and scalability

Topology	Fault tolerance	Scalability
Star	Medium	Good
Mesh	High	Excellent
Hybrid	High	Hard

Read this table as a set of trade-offs rather than a ranking. Nothing scores well on everything, which is precisely why hybrid designs exist and why "which topology is best?" is not a well-formed question. The useful question is always: best for what, at what size, and at what cost?

4. Hardware meets software: the cloud network

4.1 The evolutionary shift

Traditional networking relies on physical boxes, specialised cables and manual configuration. In the modern cloud era the logic is unchanged — the same layers, the same devices, the same topologies — but the execution is purely software-defined.

Traditional	Becomes	Cloud era
Physical box	→	Virtual instance
Copper cables	→	Virtualised overlays
Manual wiring	→	API calls or console clicks

The reason this matters for your learning: none of the previous three sections become obsolete in the cloud. A load balancer in AWS is still a load balancer. A subnet still partitions a Layer 3 network. You are learning the concepts; the cloud changes only how you instantiate them.

4.2 The AWS foundation: regions and edge locations

- **Regions.** Isolated geographic areas — Mumbai, for instance — each containing multiple data centre clusters. There are 30 or more globally, joined by an ultra-low-latency backbone.
- **Edge locations.** CDN endpoints that cache content physically closer to the user in order to reduce latency.

Edge locations are Lecture 2 made concrete. Caching content closer to the user shortens d in d/s , and propagation delay falls accordingly. This is the same reasoning that has Netflix placing appliances inside ISP networks.

4.3 Redundancy: availability zones

- **What an AZ is.** An Availability Zone consists of one or more discrete data centres with redundant power, networking and connectivity — an isolated fault domain.
- **Why you use more than one.** Deploying across two or more AZs means your application survives a fire, a flood or a power grid failure at any one site.

4.4 The Virtual Private Cloud

- **Logical isolation.** A VPC is your own private, software-defined network inside the AWS cloud. You control the IP address ranges, the subnets and the routing.
- **Subnets and AZs.** Subnets map to specific Availability Zones, partitioning your Layer 3 network for both scale and security.

Addressing in a VPC is expressed in CIDR notation, as demonstrated in class. That notation is the subject of upcoming work on IP addressing — for now, the idea to hold is that you are handed a block of addresses and you decide how to carve it up.
